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SECTION B

Answer **all** questions.

7. (a) Nitrobenzene can be made by the nitration of benzene.

(i) State the reagent(s) used for this reaction. [1]

....

(ii) State the **formula** of the nitrogen-containing electrophile that takes part in this reaction. [1]

....

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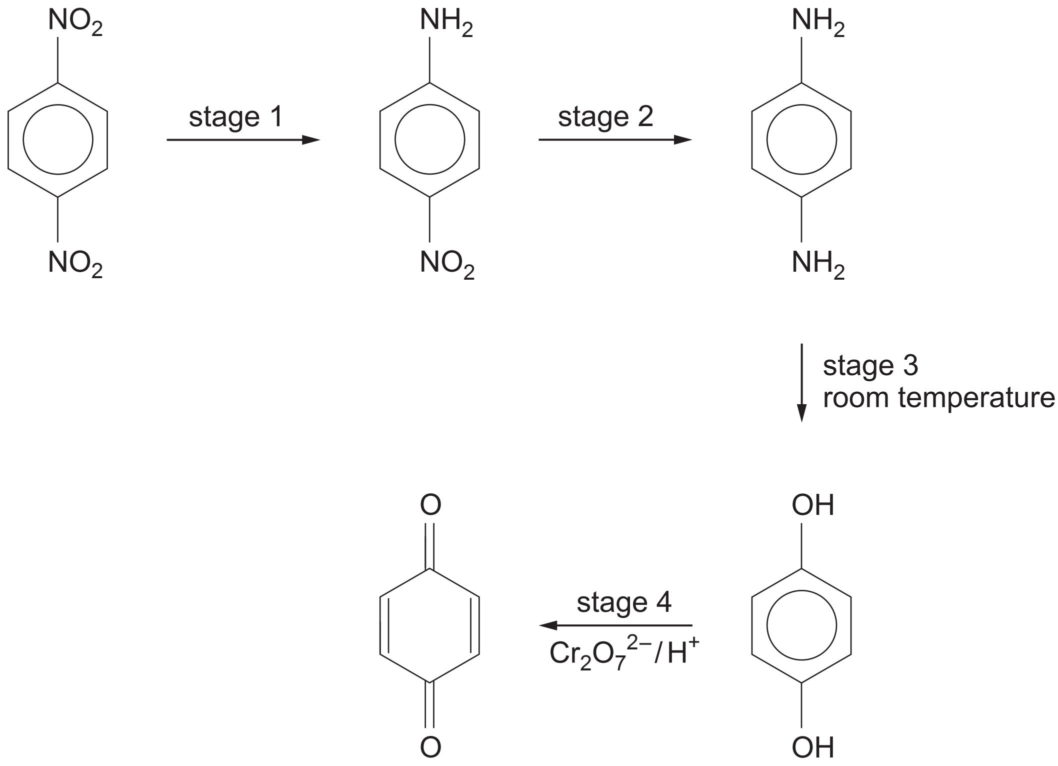
(b) Aromatic nitro-compounds are useful in synthesis as they are often the starting materials for producing other compounds.

Study the reaction sequence below and then answer the questions that follow.



cyclohexadiene-1,4-dione benzene-1,4-diol

(i) State the type of reaction occurring during stage 1. [1]

....

(ii) State the reagent(s) used in stage 2. [1]

....

(iii) State the reagent(s) used to produce benzene-1,4-diol in stage 3. [1]

....

(iv) State the colour change undergone by the reagents in stage 4. [1]

....

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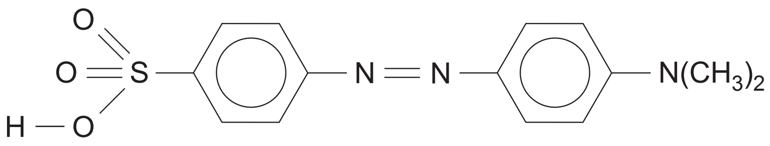
(v) The reaction product in stage 4, cyclohexadiene-1,4-dione acts as an alkene.

I. State the type of reaction mechanism that occurs when an alkene reacts with bromine. [1]

....

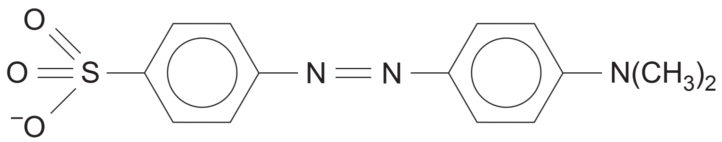
II. Draw the structure of the compound obtained when 1 mol of cyclohexadiene-1,4-dione reacts with 2 mol of bromine. [1]

(c) Methyl orange is a water soluble acid-base indicator.



(i) Give the structure of the starting sulfur-containing compound that is coupled with N,N-dimethylphenylamine to give methyl orange. [1]

(ii) Suggest why, in basic solution, methyl orange exists as the species below. [1]



....

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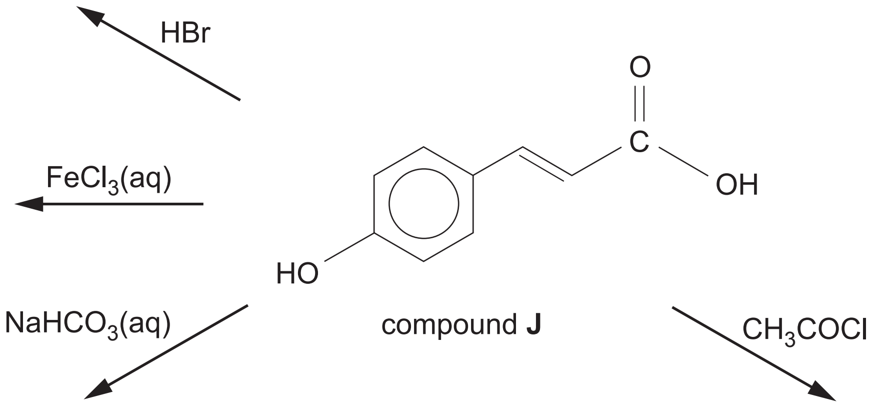
(d) Compound **J** is present in relatively large quantities in red peppers.

Study the diagram below and then answer the questions that follow.



compound **J**

(i) State what is seen when compound **J** reacts with iron(III) chloride solution. [1]

....

(ii) State what is seen when compound **J** reacts with sodium hydrogencarbonate solution. [1]

....

(iii) Give the structure of the compound formed when compound **J** reacts with hydrogen bromide. [1]

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(iv) Give the structure of the compound formed when compound **J** reacts with ethanoyl chloride. [1]

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